Chapter - 8

Candidate Key
↳ Minimal super key.

ID, name, address → Super Key

ID.
ID, name
ID, name, address → Super Key

ID, name → Not Candidate Key
ID → Candidate Key

A	B	C
1	x	y
2	a	b
3	6	d

Functional Dependency $A \rightarrow BC$

→ (stu_id, stu_name, stu_address, course_name, prereq_name, marks, grade)

Functional Dependency
ID → name, address
Course → prereq
marks → grade

$A \rightarrow B$ ✗
If A is repeated
B should have same value
$B \rightarrow A$ ✓

A	B
1	3
1	5
2	6
3	7

course & pre req

$A \rightarrow B$ ✓
$B \rightarrow A$ ✓

	A	B
T_1	1	3
T_2	1	3
	2	6
	3	7

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* Functional Dependency

Let R be a relational Schema where $\alpha \subset R$, $\beta \subset R$ then Functional Dependency $\alpha \rightarrow \beta$ holds if and only if any two tuple T_1 & T_2 aggree on α they must agree on beta (β)

→ Purpose of Functional Dependency
— To identify redundancy & reduce it

→ An Attribute is not required to be a Key for functional dependency

* Closure Set of Functional Dependency.

Set of all Functional Dependency on a Relation → Closure Set

Set F of Functional Dependency

find → $\boxed{F^+}$ (Additional Functional Dependency)

$A \rightarrow B$
$B \rightarrow C$ $\Rightarrow A \rightarrow C$

$A \rightarrow BC$
$A \rightarrow B$
$A \rightarrow C$

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* Armstrong's Axioms

$\alpha \Rightarrow$ Set of Attributes.

$\beta \subseteq \alpha$

$\alpha \rightarrow \beta$ holds

(1) Reflexivity

$\underset{\alpha}{AB} \rightarrow \underset{\beta}{A}$ (If you know A, you know A you need not determine A (of β))

$BA \rightarrow B$

$\alpha \rightarrow \beta$

α determines β

(2) Augmentation.

if $\alpha \rightarrow \beta$ holds, γ (another attribute)

$\gamma\alpha \rightarrow \gamma\beta$ holds.

(3) Transitivity.

If $\alpha \rightarrow \beta$ holds,
$\beta \rightarrow \gamma$ holds

then
$\alpha \rightarrow \gamma$, will hold.

(4) Union.

If $\alpha \rightarrow \beta$ holds
$\alpha \rightarrow \gamma$ holds,

then, $\alpha \rightarrow \beta\gamma$ will hold.

5. Decomposition

If $\alpha \rightarrow \beta\gamma$ holds
then $\alpha \rightarrow \beta$ &
$\alpha \rightarrow \gamma$ holds

6. Pseudo Transitivity

If $\alpha \rightarrow \beta$ holds,
& $\gamma\beta \rightarrow \delta$ holds,
then
$\alpha\gamma \rightarrow \delta$ will hold.

Ex –

$A \rightarrow B$
$A \rightarrow C$
$CG \rightarrow H$
$CG \rightarrow I$
$B \rightarrow H$

$A \rightarrow BC$
$CG \rightarrow HI$
$A \rightarrow H$
$AG \rightarrow I$
$AG \rightarrow H$

All found $\rightarrow$ F^+
$\hookrightarrow$ Closure set

* Closure set of Attributes

$A \rightarrow B$
$B \rightarrow D$
$C \rightarrow DE$
$CD \rightarrow AB$

$A^+ = \{A, B, D\}$
$C^+ = \{C, D, E, A, B\}$
$\hookrightarrow$ Key (super)

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F.D {
$A \rightarrow B$
$A \rightarrow C$
$CG \rightarrow H$
$CG \rightarrow I$
$B \rightarrow H$
}

$AG^+ \Rightarrow AGBCHI$.

Algorithm
α^+

result = α;
repeat
for each FD $\beta \rightarrow \gamma$
begin
if $\beta \subseteq$ result
result = result $\cup\ \gamma$
end.
Until (result does not change)

result = AG
result = AG $\cup$ B = AGB
result = AGB $\cup$ C = AGBC
result = AGBC $\cup$ H = AGBCH
result = AGBCH $\cup$ I = AGBCHI

(Application of Algorithm)
Application ~~to~~ used to
- Find ~~Primary~~ Super Key (is α Super Key)

- To check Functional Dependency
$\alpha \rightarrow \beta$, holds
(if α^+ gives β then true)

Ex- Here $AG \rightarrow B$ ✓

Set of all Attribute is always
Candidate key.

- To compute F^+

* How to find Candidate Key using Closure.

R(A,B,C,D)

Start with 1 Element Closure then Upward.

$A \to B$
$B \to C$
$C \to D$
$D \to A$

$A^+ \Rightarrow ABCD$
$B^+ \Rightarrow BCDA$
$C^+ \Rightarrow CDAB$
$D^+ \Rightarrow DABC.$

Don't check Further

A, B, C, D are Candidate
All other are Super Key
they can not be Candidate

→ R(ABCDEF)

(i)	CD	$C \to F$
~~(ii)~~	EC	$E \to A$
(iii)	AE	$EC \to D$
(iv)	AC	$A \to B$

$CD^+ \Rightarrow CDF$.

$EC^+ \Rightarrow ECDABF$ $\longrightarrow$ $E^+ \Rightarrow EAB$

$C^+ \Rightarrow CF$.

$AE^+ \Rightarrow AEB$. None of them is super Key

$AC^+ \Rightarrow ACBF$

so EC is candidate Key.

Short Cut

$X \Rightarrow (ABCDEF)$ $\longrightarrow$ on Right side

So C, E cannot be derived
so they should be on Left side
so check

CE^+ then
C^+
E^+

You get your Answer

Q. $R(CEFGHIJKLMN)$

$EF \rightarrow G$.
$EH \rightarrow KL$
$F \rightarrow IJ$
$K \rightarrow M$
$L \rightarrow N$

$E^+ \Rightarrow E$. $EH^+ \Rightarrow EHKLMN$

$F^+ \Rightarrow FIJ$ $EF^+ \Rightarrow EFGIJ$

$H^+ \Rightarrow H$ $HF^+ \Rightarrow HFIJ$

$EFH^+ \Rightarrow EFHGIJKLMN$

↳ Minimal Super Key

⇓

Candidate Key.

In Short Cut
Don't check for subset (for CK)

Q. $R(ABCD)$

$AB \to CD$
$D \to A$

$B^+ \Rightarrow B$ (Not sufficient B must be there.

$AB^+ \Rightarrow ABCD$
$BD^+ \Rightarrow BDAC$.

Q. $R(ABCDEF)$

$AB \to C$
$C \to D$
$B \to AE$

$BF^+ \Rightarrow BFAECD$.

BF is .

Q. $R(ABCD)$

$AB \to CD$
$C \to A$
$D \to B$

AB is Candidate Key
(~~Shortcut~~ not possible here)